Sabre Industries™
INNOVATION DELIVERED

Structural Design Report

180' Monopole
Site: US-NY-2029, NY

Prepared for: PTI SERVICES
by: Sabre Industries ™

Job Number: 25-3277-JDS-R1
Revision B

July 15, 2025

STATE OF NEW YORK
ROBERT E. BEACOM
090088
LICENSED PROFESSIONAL ENGINEER

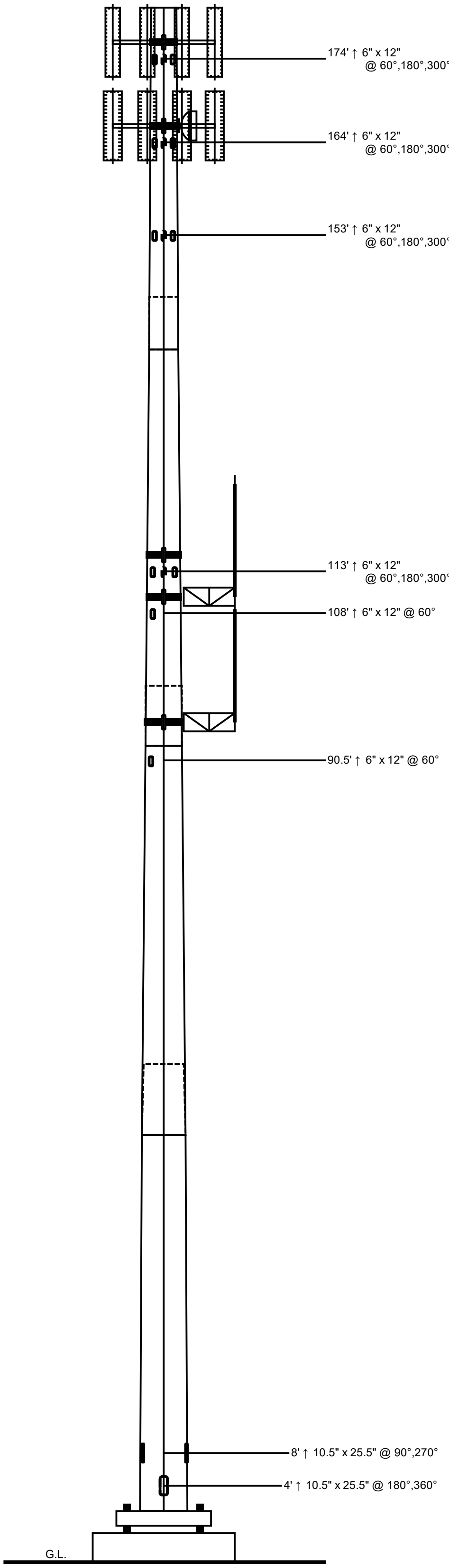
7/15/25

DATE OF RENEWAL: 7/31/23
DATED OF EXPIRATION: 7/31/26

Robert Beacom

Digitally signed by Robert Beacom
DN: c=US, serialNumber=MAS20250205984402, st=Texas, l=Alvarado, 2.5.4.97=NTRUS+DE-4349737, o=SABRE INDUSTRIES, INC., cn=Robert Beacom
Date: 2025.07.15 14:22:51 -05'00'

Section	Length (ft)	Number Of Sides	Thickness (in)	Lap Splice (ft)	Top Diameter (in)	Bottom Diameter (in)	Taper (in/ft)	Grade	Weight (lbs)	Overall Steel Height (ft)
4	40'-9"	18	5/16"		36"	43.84"	0.1925	A572-65	6041	179
				6' - 3"						
3	53'-6"	18	5/16"		42.02"	52.32"	0.1925	A572-65	9106	
				7' - 3"						
2	53'-6"	18	3/8"		50.29"	60.59"	0.1925	A572-65	12697	
				8' - 6"						
1	53'-3"	18	3/8"		58.21"	68.46"	0.1925	A572-65	16267	



Design Criteria - ANSI/TIA-222-H

Wind Speed (No Ice)	109 mph
Wind Speed (Ice)	40 mph
Design Ice Thickness	1.50 in
Risk Category	II
Exposure Category	C
Topographic Factor Procedure	Method 1 (Simplified)
Topographic Category	1
Ground Elevation	435 ft
Seismic Importance Factor, Ie	1.00
0.2-sec Spectral Response, Ss	0.142 g
1-sec Spectral Response, S1	0.051 g
Site Class	D (DEFAULT)
Seismic Design Category	B
Basic Seismic Force-Resisting System	Telecommunication Tower (Pole: Steel)

Limit State Load Combination Reactions

Load Combination	Axial (kips)	Shear (kips)	Moment (ft-k)	Deflection (ft)	Sway (deg)
1.2 D + 1.0 Wo	69.92	48.07	6352.75	9.54	4.99
0.9 D + 1.0 Wo	52.63	48.67	6329.41	9.42	4.92
1.2 D + 1.0 Di + 1.0 Wi	117.18	11.88	1620.51	2.47	1.29
1.2 D + 1.0 Ev + 1.0 Eh	71.69	1.73	258.43	0.4	0.21
0.9 D - 1.0 Ev + 1.0 Eh	50.69	1.72	254.23	0.4	0.21
1.0 D + 1.0 Wo (Service @ 60 mph)	58.22	13.08	1729.19	2.61	1.36

Base Plate Dimensions

Shape	Diameter	Thickness	Bolt Circle	Bolt Qty	Bolt Diameter
Round	81.25"	2"	75.5"	18	2.25"

Anchor Bolt Dimensions

Length	Diameter	Hole Diameter	Weight	Type	Finish
84"	2.25"	2.625"	2179.8	A615-75	Galv

Notes

1) Antenna Feed Lines Run Inside Pole
2) All dimensions are above ground level, unless otherwise specified.
3) Weights shown are estimates. Final weights may vary.
4) Full Height Step Bolts
5) Tower Rating: 97.5%
6) This tower design and, if applicable, the foundation design(s) shown on the following page(s) also meet or exceed the requirements of the 2020 New York State Building Code.

Sabre Industries™ INNOVATION DELIVERED

Sabre Industries
7101 Southbridge Drive
P.O. Box 658
Sioux City, IA 51102-0658
Phone: (712) 258-6690
Fax: (712) 279-0814

Job:	25-3277-JDS-R1-RB
Customer:	PTI SERVICES
Site Name:	US-NY-2029, NY
Description:	180' Monopole
Date:	7/15/2025 By: REB

Designed Appurtenance Loading

Elev	Description	Tx-Line
176	SitePro1 F4P-12W	
176	(8) NNH4-65C-R6	(9) 1"
176	(4) AIR 6472 B77G B77M	
176	(4) RRU 4890	
176	(4) RRU 4490	
176	(4) RRU (17" x 18" x 7")	
176	(3) DC9-48-60-24-8C-EV	(3) 1/2"
166	(1) Dish Mount (Monopole Only) - Pipe Mount (up to 6' Dish)	
166	SitePro1 F4P-12W	
166	(4) AIR 6419 B41	(4) 6x24 Hybrid
166	(4) RRUS 4460 B25+B66	(2) 1 1/2"
166	(4) RRUS 4480 B71+B85	
166	(2) IP-20C	(2) 1/4"
166	(4) 840590966	

Elev	Description	Tx-Line
166	(1) VHLP3-11W	
155	(1) 250 Sq. Ft. EPA (8000 lbs)	(6) 1 5/8"
116.72	(1) SC46A-HF1LDF(DXX-PIP)	(1) 7/8"
115	(3) Armor Tower Arched Boom - 8' Face	
115	(6) NHH-55C-R2B	
115	(1) RVZDC-6627-PF-48	
115	(1) RxxDC-3315-PF-48	(3) 1-5/8 Hybrid
115	(3) RF4461d-13a	
115	(3) RF4801d-25B	
115	(3) MT6413-77A	
110	6ft Sidearm	
110	(1) TTA (31.25" x 13.25" x 11.5")	(1) 1/2"
101.72	(1) SC46A-HF1LDF(DXX-PIP)	(1) 7/8"
95	6ft Sidearm	

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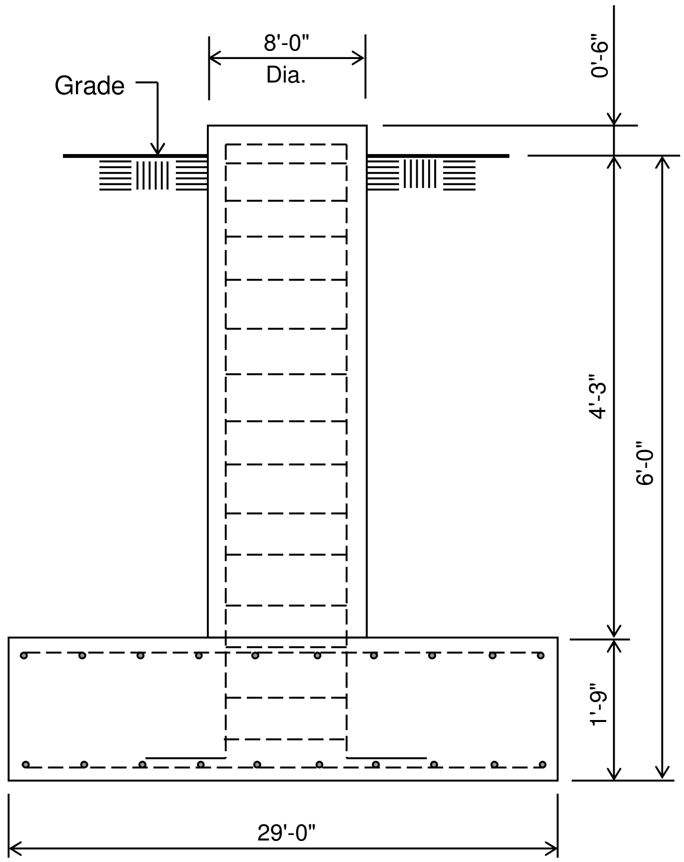
Site: US-NY-2029, NY
180’ Monopole



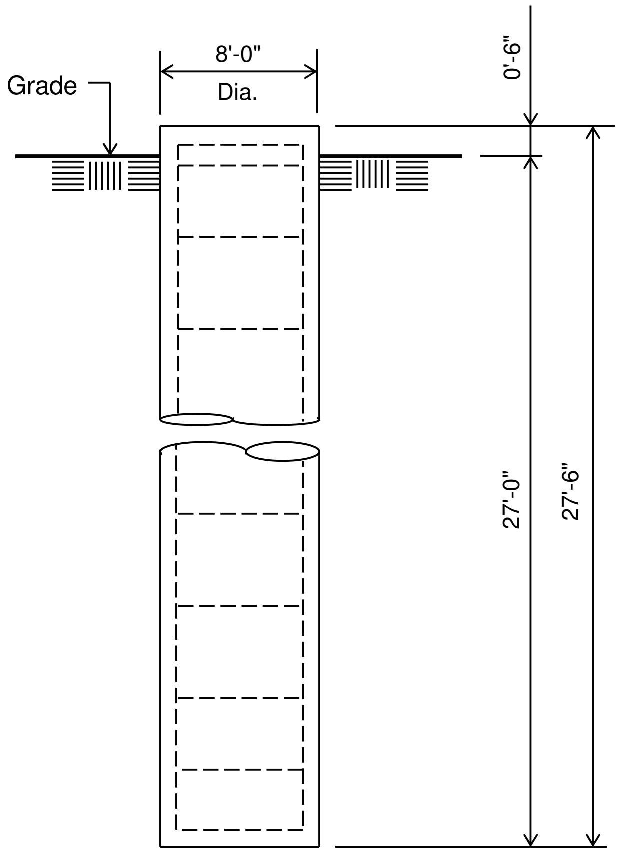
ELEVATION VIEW
(63.35 Cu. Yds.)
(1 REQUIRED; NOT TO SCALE)

Notes:

1) Concrete shall have a minimum 28-day compressive strength of 4,500 psi, in accordance with ACI 318-14.
2) Rebar to conform to ASTM specification A615 Grade 60.
3) All rebar to have a minimum of 3” concrete cover.
4) All exposed concrete corners to be chamfered 3/4”.
5) The foundation design is based on the geotechnical report by Delta Oaks Group, Project No. GEO24-23357-08, dated November 17, 2024.
6) See the geotechnical report for compaction requirements, if specified.
7) 4.25 ft of soil cover is required over the entire area of the foundation slab.
8) The bottom anchor bolt template shall be positioned as closely as possible to the bottom of the anchor bolts.

Rebar Schedule for Pad and Pier	
Pier	(50) #8 vertical rebar w/ hooks at bottom w/ #5 ties, (2) within top 5" of pier, then 4" C/C
Pad	(51) #10 horizontal rebar evenly spaced each way top and bottom (204 total)

Customer: PTI SERVICES
Site: US-NY-2029, NY
180' Monopole



ELEVATION VIEW
(51.20 Cu. Yds.)
(1 REQUIRED; NOT TO SCALE)

Notes:

1) Concrete shall have a minimum 28-day compressive strength of 4,500 psi, in accordance with ACI 318-14.
2) Rebar to conform to ASTM specification A615 Grade 60.
3) All rebar to have a minimum of 3" concrete cover.
4) All exposed concrete corners to be chamfered 3/4".
5) The foundation design is based on the geotechnical report by Delta Oaks Group, Project No. GEO24-23357-08, dated November 17, 2024.
6) See the geotechnical report for drilled pier installation requirements, if specified.
7) The bottom anchor bolt template shall be positioned as closely as possible to the bottom of the anchor bolts.

Rebar Schedule for Pier	
Pier	(52) #8 vertical rebar w/ #5 ties, (2) within top 5" of pier, then 7" C/C

7101 Southbridge Drive - P.O. Box 658 - Sioux City, IA 51102-0658 - Phone 712.258.6690 - Fax 712.279.0814

180' Monopole / US-NY-2029, NY

* All pole diameters shown on the following pages are across corners.
See profile drawing for widths across flats.

POLE GEOMETRY

ELEV ft	SECTION NAME	No. SIDE	OUTSIDE DIAM in	THICK-NESS in	RESISTANCES •*Pn kip	RESISTANCES •*Mn ft-kip	SPLICE TYPE	...OVERLAP... LENGTH ft	...OVERLAP... RATIO	w/t
179.0										
			36.56	0.312	2508.9	1840.0				
	A	18								19.3
			43.29	0.312	2808.8	2446.1				
144.5										
			43.29	0.312	2808.8	2446.1				
	A/B	18					SLIP	6.25	1.73	
			43.90	0.312	2833.3	2502.5				
138.2										
			43.90	0.312	2833.3	2502.5				
	B	18								23.3
			51.69	0.312	3110.1	3242.1				
98.2										
			51.69	0.312	3110.1	3242.1				
	B/C	18					SLIP	7.25	1.68	
			52.50	0.375	4063.8	4292.4				
91.0										
			52.50	0.375	4063.8	4292.4				
	C	18								23.4
			59.85	0.375	4386.1	5291.6				
53.2										
			59.85	0.375	4386.1	5291.6				
	C/D	18					SLIP	8.50	1.70	
			60.78	0.375	4422.1	5418.4				
44.7										
			60.78	0.375	4422.1	5418.4				
	D	18								27.2
			69.51	0.375	4714.7	6617.7				
0.0										

POLE ASSEMBLY

SECTION NAME	BASE ELEV ft	BOLTS AT BASE OF SECTION NUMBER	TYPE	DIAM in	STRENGTH ksi	THREADS IN SHEAR PLANE	CALC BASE ELEV ft
A	138.250	0	A325	0.00	92.0	0	138.250
B	91.000	0	A325	0.00	92.0	0	91.000
C	44.750	0	A325	0.00	92.0	0	44.750
D	0.000	0	A325	0.00	92.0	0	0.000

POLE SECTIONS

SECTION NAME	No.of SIDES	LENGTH ft	OUTSIDE.DIAMETER BOT * in	OUTSIDE.DIAMETER TOP * in	BEND RAD in	MAT-ERIAL ID	FLANGE.ID BOT	FLANGE.ID TOP	FLANGE.WELD ..GROUP.ID.. BOT	FLANGE.WELD ..GROUP.ID.. TOP
A	18	40.75	44.52	36.56	0.625	1	0	0	0	0
B	18	53.50	53.12	42.66	0.625	2	0	0	0	0
C	18	53.50	61.53	51.07	0.625	3	0	0	0	0
D	18	53.25	69.51	59.10	0.625	4	0	0	0	0

* - Diameter of circumscribed circle

MATERIAL TYPES
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TYPE OF SHAPE	TYPE NO	NO OF ELEM.	ORIENT & deg	HEIGHT in	WIDTH in	.THICKNESS. WEB in	FLANGE in	IRREGULARITY .PROJECTION. % OF AREA	ORIENT deg
PL	1	1	0.0	44.52	0.31	0.312	0.312	0.00	0.0
PL	2	1	0.0	53.12	0.31	0.312	0.312	0.00	0.0
PL	3	1	0.0	61.53	0.38	0.375	0.375	0.00	0.0
PL	4	1	0.0	69.51	0.38	0.375	0.375	0.00	0.0

& - With respect to vertical

MATERIAL PROPERTIES
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MATERIAL TYPE NO.	ELASTIC MODULUS ksi	UNIT WEIGHT pcf	.. STRENGTH .. Fu ksi	Fy ksi	THERMAL COEFFICIENT /deg
1	29000.0	490.0	80.0	65.0	0.00001170
2	29000.0	490.0	80.0	65.0	0.00001170
3	29000.0	490.0	80.0	65.0	0.00001170
4	29000.0	490.0	80.0	65.0	0.00001170

* Only 5 condition(s) shown in full
* RRUs/TMAs were assumed to be behind antennas

* Some concentrated wind loads may have been derived from full-scale wind tunnel testing

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LOADING CONDITION A ==

109 mph wind with no ice. Wind Azimuth: 0• (1.2 D + 1.0 Wo)

LOADS ON POLE
=============

LOAD TYPE	ELEV ft	APPLY..LOAD..AT RADIUS ft	AZI	LOAD AZI	......FORCES...... HORIZ kip	DOWN kip	......MOMENTS..... VERTICAL ft-kip	TORSNAL ft-kip
C	175.000	0.00	0.0	0.0	0.0000	1.2726	0.0000	0.0000
C	175.000	0.00	0.0	0.0	6.2439	2.4118	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0293	0.0151	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.6909	0.0000	0.0000
C	165.000	0.00	0.0	0.0	5.3090	1.3299	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0321	0.0168	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0317	0.0168	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	1.1532	0.0000	0.0000
C	154.000	0.00	0.0	0.0	10.8563	9.6000	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0313	0.0168	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0308	0.0168	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0303	0.0168	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0750	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.1731	0.0342	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0298	0.0168	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.7387	0.0000	0.0000
C	114.000	0.00	0.0	0.0	4.5371	1.8970	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0523	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1394	0.0600	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.5037	0.8488	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0292	0.0168	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0653	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.1682	0.0342	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0286	0.0168	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.4884	0.8488	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0279	0.0168	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0272	0.0168	0.0000	0.0000

C	65.000	0.00	0.0	0.0	0.0264	0.0168	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0255	0.0168	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0244	0.0168	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0232	0.0168	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0216	0.0168	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0194	0.0168	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0870	0.1466	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0968	0.1695	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0983	0.3453	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0983	0.3453	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0988	0.1761	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1070	0.2027	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1079	0.4539	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.4539	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.2520	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1100	0.2771	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1096	0.2821	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1096	0.2821	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1091	0.5738	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1091	0.5738	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1082	0.2917	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0946	0.3178	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0948	0.3230	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0963	0.3282	0.0000	0.0000

ANTENNA LOADING

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.......ANTENNA.................			ATTACHMENT		..........ANTENNA FORCES......			
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.47	0.00	0.14	0.00

==

LOADING CONDITION M ==

109 mph wind with no ice. Wind Azimuth: 0• (0.9 D + 1.0 Wo)

LOADS ON POLE

=============

LOAD	ELEV	APPLY..LOAD..AT		LOAD	......FORCES......		......MOMENTS.....	
TYPE		RADIUS	AZI	AZI	HORIZ	DOWN	VERTICAL	TORSNAL
	ft	ft			kip	kip	ft-kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0000	0.9544	0.0000	0.0000
C	175.000	0.00	0.0	0.0	6.2439	1.8088	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0293	0.0113	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.2682	0.0000	0.0000
C	165.000	0.00	0.0	0.0	5.3090	0.9974	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0321	0.0126	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0317	0.0126	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	0.8649	0.0000	0.0000
C	154.000	0.00	0.0	0.0	10.8563	7.2000	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0313	0.0126	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0308	0.0126	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0303	0.0126	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0562	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.1731	0.0257	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0298	0.0126	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.5540	0.0000	0.0000
C	114.000	0.00	0.0	0.0	4.5371	1.4227	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0392	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1394	0.0450	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.5037	0.6366	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0292	0.0126	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0489	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.1682	0.0257	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0286	0.0126	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.4884	0.6366	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0279	0.0126	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0272	0.0126	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0264	0.0126	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0255	0.0126	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0244	0.0126	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0232	0.0126	0.0000	0.0000

C	25.000	0.00	0.0	0.0	0.0216	0.0126	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0194	0.0126	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0870	0.1099	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0968	0.1271	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0983	0.2589	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0983	0.2589	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0988	0.1321	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1070	0.1520	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.1079	0.3404	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.3404	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.1079	0.1890	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1100	0.2078	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.1096	0.2116	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1096	0.2116	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.1091	0.4303	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1091	0.4303	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.1082	0.2188	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0946	0.2383	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0948	0.2422	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0963	0.2461	0.0000	0.0000

ANTENNA LOADING
===============

.......ANTENNA.................			ATTACHMENT		..........ANTENNA FORCES......			
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.47	0.00	0.10	0.00

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LOADING CONDITION Y ===

40 mph wind with 1.5 ice. Wind Azimuth: 0• (1.2 D + 1.0 Di + 1.0 Wi)

LOADS ON POLE
=============

LOAD	ELEV	APPLY..LOAD..AT		LOAD	......FORCES......		......MOMENTS.....	
TYPE		RADIUS	AZI	AZI	HORIZ	DOWN	VERTICAL	TORSNAL
	ft	ft			kip	kip	ft-kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0000	1.2726	0.0000	0.0000
C	175.000	0.00	0.0	0.0	1.4926	6.9462	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0315	0.0271	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.6909	0.0000	0.0000
C	165.000	0.00	0.0	0.0	1.2119	3.8298	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0344	0.0288	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0338	0.0288	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	1.1532	0.0000	0.0000
C	154.000	0.00	0.0	0.0	2.4860	23.6076	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0331	0.0288	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0324	0.0288	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0317	0.0288	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0750	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0465	0.0827	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0309	0.0288	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.7387	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.9593	7.6184	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0523	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0256	0.1683	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1307	1.1871	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0301	0.0288	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0653	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0448	0.0820	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0292	0.0288	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.1257	1.1822	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0282	0.0288	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0272	0.0288	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0261	0.0288	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0248	0.0288	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0234	0.0288	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0217	0.0288	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0196	0.0288	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0169	0.0288	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0224	0.2308	0.0000	0.0000

D	144.500	0.00	180.0	0.0	0.0246	0.2644	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0249	0.4421	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0249	0.4421	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0250	0.2737	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0268	0.3112	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0270	0.5641	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0270	0.5641	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0269	0.3628	0.0000	0.0000
D	64.036	0.00	180.0	0.0	0.0273	0.3880	0.0000	0.0000
D	64.036	0.00	180.0	0.0	0.0272	0.3941	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0271	0.4000	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0270	0.6927	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0270	0.6927	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0267	0.4109	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0232	0.4341	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0232	0.4348	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0234	0.4318	0.0000	0.0000

ANTENNA LOADING
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.......ANTENNA.................			ATTACHMENT		...........ANTENNA FORCES......			
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.08	0.00	0.51	0.00

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LOADING CONDITION AK ===

Seismic - Azimuth: 0• (1.2 D + 1.0 Ev + 1.0 Eh)

LOADS ON POLE
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LOAD	ELEV	APPLY..LOAD..AT		LOAD	......FORCES......		......MOMENTS.....	
TYPE		RADIUS	AZI	AZI	HORIZ	DOWN	VERTICAL	TORSNAL
	ft	ft			kip	kip	ft-kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0751	1.3046	0.0000	0.0000
C	175.000	0.00	0.0	0.0	0.1423	2.4725	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0009	0.0155	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0887	1.7335	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0186	0.3629	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0697	1.3635	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0009	0.0172	0.0000	0.0000
C	158.620	0.00	0.0	0.0	0.3160	6.6850	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0008	0.0172	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.4386	9.8416	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0527	1.1822	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0007	0.0172	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0006	0.0172	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0005	0.0172	0.0000	0.0000
C	117.750	0.00	0.0	0.0	0.2705	10.3817	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0009	0.0351	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0019	0.0769	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0004	0.0172	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0475	1.9447	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0185	0.7573	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0014	0.0615	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0012	0.0536	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0194	0.8702	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0004	0.0172	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0007	0.0351	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0013	0.0669	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0003	0.0172	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.0144	0.8702	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0002	0.0172	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0002	0.0172	0.0000	0.0000
C	71.500	0.00	0.0	0.0	0.1407	14.6419	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0001	0.0172	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0001	0.0172	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0001	0.0172	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0000	0.0172	0.0000	0.0000
C	26.620	0.00	0.0	0.0	0.0222	16.6612	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0000	0.0172	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0000	0.0172	0.0000	0.0000

D	179.000	0.00	180.0	180.0	0.0000	0.0000	0.0000	0.0000
D	0.000	0.00	180.0	180.0	0.0000	0.0000	0.0000	0.0000

ANTENNA LOADING

.......ANTENNA.................			ATTACHMENT		..........ANTENNA FORCES......			
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.00	0.00	0.00	0.00

LOADING CONDITION AL

Seismic - Azimuth: 0• (0.9 D - 1.0 Ev + 1.0 Eh)

LOADS ON POLE

LOAD	ELEV	APPLY..LOAD..AT		LOAD	......FORCES......		......MOMENTS.....	
TYPE		RADIUS	AZI	AZI	HORIZ	DOWN	VERTICAL	TORSNAL
	ft	ft			kip	kip	ft-kip	ft-kip
C	175.000	0.00	0.0	0.0	0.0751	0.9224	0.0000	0.0000
C	175.000	0.00	0.0	0.0	0.1423	1.7481	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0009	0.0109	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0887	1.2256	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0186	0.2566	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0697	0.9640	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0009	0.0122	0.0000	0.0000
C	158.620	0.00	0.0	0.0	0.3160	4.7266	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0008	0.0122	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.4386	6.9584	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0527	0.8359	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0007	0.0122	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0006	0.0122	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0005	0.0122	0.0000	0.0000
C	117.750	0.00	0.0	0.0	0.2705	7.3402	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0009	0.0248	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0019	0.0544	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0004	0.0122	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0475	1.3750	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0185	0.5354	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0014	0.0435	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0012	0.0379	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0194	0.6152	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0004	0.0122	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0007	0.0248	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0013	0.0474	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0003	0.0122	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.0144	0.6152	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0002	0.0122	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0002	0.0122	0.0000	0.0000
C	71.500	0.00	0.0	0.0	0.1407	10.3525	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0001	0.0122	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0001	0.0122	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0001	0.0122	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0000	0.0122	0.0000	0.0000
C	26.620	0.00	0.0	0.0	0.0222	11.7801	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0000	0.0122	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0000	0.0122	0.0000	0.0000
D	179.000	0.00	180.0	180.0	0.0000	0.0000	0.0000	0.0000
D	0.000	0.00	180.0	180.0	0.0000	0.0000	0.0000	0.0000

ANTENNA LOADING

.......ANTENNA.................			ATTACHMENT		..........ANTENNA FORCES......			
TYPE	ELEV	AZI	RAD	AZI	AXIAL	SHEAR	GRAVITY	TORSION
	ft		ft		kip	kip	kip	ft-kip
HP	165.0	0.0	2.3	0.0	0.00	0.00	0.00	0.00

180' Monopole / US-NY-2029, NY

MAXIMUM POLE DEFORMATIONS CALCULATED(w.r.t. wind direction)

MAST ELEV ft	DEFLECTIONS (ft) HORIZONTAL ALONG	ACROSS	DOWN	ROTATIONS (deg) TILT ALONG	ACROSS	TWIST
179.0	9.54A	-0.19Q	0.65A	4.99A	-0.10Q	-0.01U
174.1	9.12A	-0.18Q	0.61A	4.99A	-0.10Q	-0.01U
169.1	8.69A	-0.18Q	0.58A	4.98A	-0.10Q	-0.01U
164.2	8.26A	-0.17Q	0.54A	4.97A	-0.10Q	-0.01U
159.3	7.84A	-0.16Q	0.50A	4.95A	-0.10Q	-0.01U
154.4	7.42A	-0.15Q	0.47A	4.91A	-0.10Q	-0.01U
149.4	7.00A	-0.14Q	0.43A	4.87A	-0.10Q	-0.01U
144.5	6.59A	-0.13Q	0.40A	4.80A	-0.10Q	-0.01U
138.2	6.07A	-0.12Q	0.35A	4.70A	-0.10Q	-0.01U
132.5	5.62A	-0.11Q	0.32A	4.58A	-0.09Q	-0.01U
126.8	5.17A	-0.10Q	0.28A	4.44A	-0.09Q	-0.01U
121.1	4.73A	-0.09Q	0.25A	4.29A	-0.09Q	-0.01U
115.4	4.32A	-0.09Q	0.22A	4.13A	-0.08Q	-0.01U
109.7	3.92A	-0.08Q	0.19A	3.96A	-0.08Q	-0.01U
104.0	3.53A	-0.07Q	0.16A	3.77A	-0.08Q	-0.01U
98.2	3.17A	-0.06Q	0.14A	3.57A	-0.07Q	-0.01U
91.0	2.73A	-0.05Q	0.11A	3.35A	-0.07Q	-0.01U
85.6	2.42A	-0.05Q	0.09A	3.18A	-0.06Q	-0.01U
80.2	2.13A	-0.04Q	0.08A	3.00A	-0.06Q	-0.01U
74.8	1.86A	-0.04Q	0.06A	2.81A	-0.06Q	-0.01U
69.4	1.61A	-0.03Q	0.05A	2.62A	-0.05Q	-0.01U
64.0	1.37A	-0.03Q	0.04A	2.43A	-0.05Q	0.00U
58.6	1.15A	-0.02Q	0.03A	2.23A	-0.04Q	0.00U
53.2	0.95A	-0.02Q	0.02A	2.04A	-0.04Q	0.00U
44.7	0.67A	-0.01Q	0.01A	1.72A	-0.03Q	0.00U
39.2	0.51A	-0.01Q	0.01A	1.51A	-0.03Q	0.00U
33.6	0.38A	-0.01Q	0.01A	1.29A	-0.03Q	0.00U
28.0	0.26A	0.00Q	0.00A	1.08A	-0.02Q	0.00U
22.4	0.17A	0.00Q	0.00A	0.86A	-0.02Q	0.00U
16.8	0.09A	0.00Q	0.00A	0.64A	-0.01Q	0.00U

11.2	0.04A	0.00Q	0.00Z	0.43A	-0.01Q	0.00U
5.6	0.01A	0.00Q	0.00Z	0.21A	0.00Q	0.00U
0.0	0.00A	0.00A	0.00A	0.00A	0.00A	0.00A

MAXIMUM ANTENNA AND REFLECTOR ROTATIONS

ELEV ft	ANT AZI deg	ANT TYPE	BEAM DEFLECTIONS (deg) ROLL	YAW	PITCH	TOTAL
165.0	0.0	HP	4.937 D	0.189 K	4.974 A	4.974 A

MAXIMUM POLE FORCES CALCULATED(w.r.t. to wind direction)

MAST ELEV ft	TOTAL AXIAL kip	SHEAR.w.r.t.WIND.DIR ALONG kip	SHEAR.w.r.t.WIND.DIR ACROSS kip	MOMENT.w.r.t.WIND.DIR ALONG ft-kip	MOMENT.w.r.t.WIND.DIR ACROSS ft-kip	TORSION ft-kip
179.0						
	-0.09 A	-0.16 M	0.12 B	0.62 R	-0.37 H	-0.03 B
	9.40 AF	6.84 R	0.12 B	-7.65 B	-0.35 B	-0.04 H
174.1						
	9.41 Y	7.02 M	-0.39 O	-7.70 H	-0.29 R	-0.03 F
	10.58 Y	7.45 M	-0.39 O	-44.00 K	1.77 O	-0.19 O
169.1						
	10.58 AJ	7.92 W	0.35 W	-43.75 T	1.72 O	-0.20 O
	17.84 AJ	13.99 W	-0.42 Q	-88.96 W	2.78 O	-0.70 U
164.2						
	17.85 AJ	13.61 R	-0.62 K	-88.73 W	2.86 O	-0.69 U
	19.07 AJ	14.06 R	-0.62 K	-158.83 A	4.43 Q	0.86 E
159.3						
	19.06 AI	14.55 R	0.60 T	-158.84 A	4.62 Q	0.84 E
	20.34 AI	15.04 R	0.60 T	-233.44 R	-7.14 T	1.07 E
154.4						
	20.34 AJ	14.80 T	-0.63 K	-233.27 R	-7.32 T	1.08 E
	46.36 AJ	26.12 T	-0.63 K	-362.35 R	-10.04 T	1.40 E
149.4						
	46.36 Z	26.21 T	-0.56 Q	-362.72 R	-10.50 T	1.40 E
	47.68 Z	26.71 T	-0.56 Q	-497.99 A	-12.50 T	1.58 E
144.5						
	47.69 Z	26.63 T	-0.58 K	-497.75 A	-12.68 T	1.54 E
	50.45 Z	27.24 T	-0.58 K	-676.34 A	-15.79 T	1.90 E
138.2						
	50.45 Z	27.31 T	-0.75 Q	-676.31 A	-15.83 T	1.91 E
	52.06 Z	27.91 T	-0.75 Q	-844.01 A	18.83 Q	-2.20 U
132.5						
	52.06 Z	27.85 T	-0.58 K	-844.44 A	18.75 Q	-2.22 U
	53.67 Z	28.42 T	-0.58 K	-1015.73 A	22.22 K	-2.59 U
126.8						
	53.67 Z	28.73 T	0.56 N	-1015.76 A	22.24 K	-2.63 U
	55.34 Z	29.34 T	0.56 N	-1191.36 A	25.34 K	-3.02 U
121.1						
	55.34 Z	29.23 A	0.68 N	-1191.12 A	25.33 Q	-3.00 U
	57.16 Z	29.99 A	0.68 N	-1370.58 A	28.89 Q	-3.45 U
115.4						
	57.16 Z	29.99 A	0.76 N	-1370.43 A	28.86 Q	-3.46 U
	67.25 Z	35.15 A	0.76 N	-1574.85 A	32.73 Q	-3.91 U
109.7						

	67.25 Z	35.12 A	-0.77 Q	-1575.12 A	32.79 Q	-3.91 U
	70.42 Z	36.39 A	-0.77 Q	-1793.02 A	37.35 Q	-4.36 U
104.0						
	70.42 Z	36.39 A	-0.81 Q	-1792.91 A	37.23 Q	-4.34 U
	72.33 Z	37.16 A	-0.81 Q	-2015.10 A	42.03 Q	-4.80 U
98.2						
	72.33 Z	37.09 A	-0.88 Q	-2015.12 A	41.90 Q	-4.81 U
	77.63 Z	38.39 A	-0.88 Q	-2304.12 A	48.51 Q	-5.30 U
91.0						
	77.63 Z	38.54 U	-0.77 Q	-2303.97 A	48.69 Q	-5.29 U
	79.60 Z	39.12 U	-0.77 Q	-2523.93 A	52.99 Q	-5.70 U
85.6						
	79.60 Z	39.13 U	0.71 N	-2523.48 A	53.19 Q	-5.69 U
	81.63 Z	39.74 U	0.71 N	-2746.71 A	56.76 Q	-6.09 U
80.2						
	81.63 Z	39.77 U	-0.84 Q	-2746.83 A	56.87 Q	-6.09 U
	83.68 Z	40.38 U	-0.84 Q	-2971.50 A	61.55 Q	-6.43 U
74.8						
	83.68 Z	40.43 U	-0.80 Q	-2971.56 A	61.55 Q	-6.44 U
	85.73 Z	41.02 U	-0.80 Q	-3198.03 A	66.00 Q	-6.78 U
69.4						
	85.73 Z	41.10 U	-0.81 Q	-3198.02 A	66.12 Q	-6.78 U
	87.84 Z	41.72 U	-0.81 Q	-3427.58 A	70.64 Q	-7.09 U
64.0						
	87.84 Z	41.76 U	-0.83 Q	-3427.56 A	70.66 Q	-7.08 U
	89.97 Z	42.35 U	-0.83 Q	-3660.09 A	75.25 Q	-7.39 U
58.6						
	89.97 Z	42.35 U	-0.83 Q	-3659.90 A	75.30 Q	-7.39 U
	92.15 Z	42.96 U	-0.83 Q	-3895.55 A	79.89 Q	-7.67 U
53.2						
	92.15 Z	42.99 U	-0.82 Q	-3895.53 A	79.88 Q	-7.67 U
	98.07 Z	43.94 U	-0.82 Q	-4272.58 A	87.07 Q	-8.05 U
44.7						
	98.07 Z	43.96 U	-0.82 Q	-4272.62 A	87.07 Q	-8.05 U
	100.38 Z	44.56 U	-0.82 Q	-4524.47 A	91.76 Q	-8.27 U
39.2						
	100.38 Z	44.61 U	-0.79 Q	-4524.47 A	91.81 Q	-8.27 U
	102.74 Z	45.22 U	-0.79 Q	-4779.11 A	96.30 Q	-8.47 U
33.6						
	102.74 Z	45.27 U	-0.79 U	-4779.06 A	96.29 Q	-8.47 U
	105.09 Z	45.85 U	-0.79 U	-5036.10 A	100.19 Q	-8.65 U
28.0						
	105.09 Z	45.86 U	-0.79 U	-5036.13 A	100.19 Q	-8.65 U
	107.49 Z	46.44 U	-0.79 U	-5295.66 A	103.94 Q	-8.80 U
22.4						
	107.49 Z	46.45 U	-0.80 U	-5295.62 A	103.93 Q	-8.80 U
	109.89 Z	46.99 U	-0.80 U	-5557.39 A	107.67 Q	-8.91 U
16.8						
	109.89 Z	47.03 U	-0.79 U	-5557.39 A	107.68 Q	-8.91 U
	112.33 Z	47.58 U	-0.79 U	-5820.82 A	111.11 Q	-9.00 U
11.2						
	112.33 Z	47.59 U	-0.77 U	-5820.81 A	111.10 Q	-9.00 U
	114.76 Z	48.13 U	-0.77 U	-6085.96 A	114.60 Q	-9.05 U
5.6						
	114.76 Z	48.14 U	-0.79 U	-6085.96 A	114.59 Q	-9.05 U
	117.18 Z	48.67 U	-0.79 U	-6352.75 A	118.03 Q	-9.06 U
base reaction	117.18 Z	-48.67 U	0.79 U	6352.75 A	-118.03 Q	9.06 U

COMPLIANCE WITH 4.8.2 & 4.5.4
============================

ELEV ft	AXIAL	BENDING	SHEAR + TORSIONAL	TOTAL	SATISFIED	D/t(w/t)	MAX ALLOWED
179.00	0.00A	0.00R	0.00B	0.00R	YES	19.26A	45.2
	0.00AF	0.00B	0.01R	0.01B	YES	19.79A	45.2
174.07	0.00Y	0.00H	0.01M	0.01H	YES	19.79A	45.2
	0.00Y	0.02K	0.01M	0.02K	YES	20.33A	45.2
169.14	0.00AJ	0.02T	0.01W	0.02D	YES	20.33A	45.2
	0.01AJ	0.04W	0.01W	0.05K	YES	20.86A	45.2
164.21	0.01AJ	0.04W	0.01R	0.05K	YES	20.86A	45.2
	0.01AJ	0.07A	0.01R	0.08K	YES	21.40A	45.2
159.29	0.01AI	0.07A	0.01R	0.08A	YES	21.40A	45.2
	0.01AI	0.10R	0.01R	0.11A	YES	21.93A	45.2
154.36	0.01AJ	0.10R	0.01T	0.11A	YES	21.93A	45.2
	0.02AJ	0.15R	0.02T	0.16A	YES	22.47A	45.2
149.43	0.02Z	0.15R	0.02T	0.16A	YES	22.47A	45.2
	0.02Z	0.20A	0.02T	0.21A	YES	23.00A	45.2
144.50	0.02Z	0.20A	0.02T	0.21A	YES	23.00A	45.2
	0.02Z	0.26A	0.02T	0.27A	YES	23.68A	45.2
138.25	0.02Z	0.27A	0.02T	0.28A	YES	23.33A	45.2
	0.02Z	0.32A	0.02T	0.33A	YES	23.95A	45.2
132.54	0.02Z	0.32A	0.02T	0.33A	YES	23.95A	45.2
	0.02Z	0.37A	0.02T	0.38A	YES	24.57A	45.2
126.82	0.02Z	0.37A	0.02T	0.38A	YES	24.57A	45.2
	0.02Z	0.42A	0.02T	0.43A	YES	25.19A	45.2
121.11	0.02Z	0.42A	0.02A	0.43A	YES	25.19A	45.2
	0.02Z	0.47A	0.02A	0.48A	YES	25.81A	45.2
115.39	0.02Z	0.47A	0.02A	0.48A	YES	25.81A	45.2
	0.02Z	0.52A	0.02U	0.53A	YES	26.43A	45.2
109.68	0.02Z	0.52A	0.02A	0.53A	YES	26.43A	45.2
	0.02Z	0.57A	0.02A	0.58A	YES	27.05A	45.2
103.96	0.02Z	0.57A	0.02U	0.58A	YES	27.05A	45.2
	0.02Z	0.62A	0.02U	0.63A	YES	27.67A	45.2
98.25	0.02Z	0.48A	0.02U	0.49A	YES	23.00A	45.2
	0.02Z	0.53A	0.02U	0.54A	YES	23.66A	45.2
91.00	0.02Z	0.54A	0.02U	0.55A	YES	23.37A	45.2
	0.02Z	0.57A	0.02U	0.58A	YES	23.85A	45.2
85.61	0.02Z	0.57A	0.02U	0.58A	YES	23.85A	45.2
	0.02Z	0.60A	0.02U	0.61A	YES	24.34A	45.2
80.21	0.02Z	0.60A	0.02U	0.61A	YES	24.34A	45.2

	0.02Z	0.63A	0.02U	0.64A	YES	24.83A	45.2
74.82							
	0.02Z	0.63A	0.02U	0.64A	YES	24.83A	45.2
	0.02Z	0.66A	0.02U	0.67A	YES	25.32A	45.2
69.43							
	0.02Z	0.66A	0.02U	0.67A	YES	25.32A	45.2
	0.02Z	0.68A	0.02U	0.70A	YES	25.81A	45.2
64.04							
	0.02Z	0.68A	0.02U	0.70A	YES	25.81A	45.2
	0.02Z	0.71A	0.02U	0.72A	YES	26.29A	45.2
58.64							
	0.02Z	0.71A	0.02U	0.72A	YES	26.29A	45.2
	0.02Z	0.74A	0.02U	0.75A	YES	26.78A	45.2
53.25							
	0.02Z	0.74A	0.02U	0.75A	YES	26.78A	45.2
	0.02Z	0.77A	0.02U	0.79A	YES	27.55A	45.2
44.75							
	0.02Z	0.79A	0.02U	0.80A	YES	27.20A	45.2
	0.02Z	0.81A	0.02U	0.83A	YES	27.71A	45.2
39.16							
	0.02Z	0.81A	0.02U	0.83A	YES	27.71A	45.2
	0.02Z	0.84A	0.02U	0.85A	YES	28.21A	45.2
33.56							
	0.02Z	0.84A	0.02U	0.85A	YES	28.21A	45.2
	0.02Z	0.86A	0.02U	0.87A	YES	28.72A	45.2
27.97							
	0.02Z	0.86A	0.02U	0.87A	YES	28.72A	45.2
	0.02Z	0.88A	0.02U	0.89A	YES	29.22A	45.2
22.37							
	0.02Z	0.88A	0.02U	0.89A	YES	29.22A	45.2
	0.02Z	0.90A	0.02U	0.92A	YES	29.73A	45.2
16.78							
	0.02Z	0.90A	0.02U	0.92A	YES	29.73A	45.2
	0.02Z	0.92A	0.02U	0.94A	YES	30.24A	45.2
11.19							
	0.02Z	0.92A	0.02U	0.94A	YES	30.24A	45.2
	0.02Z	0.94A	0.02U	0.96A	YES	30.74A	45.2
5.59							
	0.02Z	0.94A	0.02U	0.96A	YES	30.74A	45.2
	0.02Z	0.96A	0.02U	0.98A	YES	31.25A	45.2
0.00							

MAXIMUM LOADS ONTO FOUNDATION(w.r.t. wind direction)

DOWN	SHEAR.w.r.t.WIND.DIR		MOMENT.w.r.t.WIND.DIR		TORSION
	ALONG	ACROSS	ALONG	ACROSS	
kip	kip	kip	ft-kip	ft-kip	ft-kip
117.18	48.67	-0.79	-6352.75	118.03	-9.06
Z	U	U	A	Q	U

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*****************************************************************************
*************************  Service Load Condition  **************************
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* Only 1 condition(s) shown in full
* RRUs/TMAs were assumed to be behind antennas

* Some concentrated wind loads may have been derived from full-scale wind tunnel testing

LOADING CONDITION A ===

60 mph wind with no ice. Wind Azimuth: 0• (1.0 D + 1.0 Wo)

LOADS ON POLE

LOAD TYPE	ELEV ft	APPLY..LOAD..AT RADIUS ft	APPLY..LOAD..AT AZI	LOAD AZI	......FORCES...... HORIZ kip	......FORCES...... DOWN kip	......MOMENTS..... VERTICAL ft-kip	......MOMENTS..... TORSNAL ft-kip
C	175.000	0.00	0.0	0.0	0.0000	1.0605	0.0000	0.0000
C	175.000	0.00	0.0	0.0	1.6928	2.0098	0.0000	0.0000
C	174.500	0.00	0.0	0.0	0.0079	0.0126	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0000	1.4091	0.0000	0.0000
C	165.000	0.00	0.0	0.0	1.4393	1.1083	0.0000	0.0000
C	165.000	0.00	0.0	0.0	0.0087	0.0140	0.0000	0.0000
C	155.000	0.00	0.0	0.0	0.0086	0.0140	0.0000	0.0000
C	154.000	0.00	0.0	0.0	0.0000	0.9610	0.0000	0.0000
C	154.000	0.00	0.0	0.0	2.9433	8.0000	0.0000	0.0000
C	145.000	0.00	0.0	0.0	0.0085	0.0140	0.0000	0.0000
C	135.000	0.00	0.0	0.0	0.0083	0.0140	0.0000	0.0000
C	125.000	0.00	0.0	0.0	0.0082	0.0140	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0000	0.0625	0.0000	0.0000
C	115.720	0.00	0.0	0.0	0.0469	0.0285	0.0000	0.0000
C	115.000	0.00	0.0	0.0	0.0081	0.0140	0.0000	0.0000
C	114.000	0.00	0.0	0.0	0.0000	0.6156	0.0000	0.0000
C	114.000	0.00	0.0	0.0	1.2300	1.5808	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0000	0.0436	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.0378	0.0500	0.0000	0.0000
C	109.000	0.00	0.0	0.0	0.1366	0.7073	0.0000	0.0000
C	105.000	0.00	0.0	0.0	0.0079	0.0140	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0000	0.0544	0.0000	0.0000
C	100.720	0.00	0.0	0.0	0.0456	0.0285	0.0000	0.0000
C	95.000	0.00	0.0	0.0	0.0078	0.0140	0.0000	0.0000
C	94.000	0.00	0.0	0.0	0.1324	0.7073	0.0000	0.0000
C	85.000	0.00	0.0	0.0	0.0076	0.0140	0.0000	0.0000
C	75.000	0.00	0.0	0.0	0.0074	0.0140	0.0000	0.0000
C	65.000	0.00	0.0	0.0	0.0072	0.0140	0.0000	0.0000
C	55.000	0.00	0.0	0.0	0.0069	0.0140	0.0000	0.0000
C	45.000	0.00	0.0	0.0	0.0066	0.0140	0.0000	0.0000
C	35.000	0.00	0.0	0.0	0.0063	0.0140	0.0000	0.0000
C	25.000	0.00	0.0	0.0	0.0059	0.0140	0.0000	0.0000
C	15.000	0.00	0.0	0.0	0.0053	0.0140	0.0000	0.0000
D	179.000	0.00	180.0	0.0	0.0236	0.1222	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0262	0.1412	0.0000	0.0000
D	144.500	0.00	180.0	0.0	0.0267	0.2877	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0267	0.2877	0.0000	0.0000
D	138.250	0.00	180.0	0.0	0.0268	0.1468	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0290	0.1689	0.0000	0.0000
D	98.250	0.00	180.0	0.0	0.0293	0.3782	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0293	0.3782	0.0000	0.0000
D	91.000	0.00	180.0	0.0	0.0292	0.2100	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.0298	0.2309	0.0000	0.0000
D	58.643	0.00	180.0	0.0	0.0297	0.2351	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0297	0.2351	0.0000	0.0000
D	53.250	0.00	180.0	0.0	0.0296	0.4781	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0296	0.4781	0.0000	0.0000
D	44.750	0.00	180.0	0.0	0.0293	0.2431	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0256	0.2648	0.0000	0.0000
D	11.188	0.00	180.0	0.0	0.0257	0.2691	0.0000	0.0000
D	0.000	0.00	180.0	0.0	0.0261	0.2735	0.0000	0.0000

ANTENNA LOADING

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.......ANTENNA................			ATTACHMENT		..........ANTENNA FORCES......			
TYPE	ELEV ft	AZI	RAD ft	AZI	AXIAL kip	SHEAR kip	GRAVITY kip	TORSION ft-kip
HP	165.0	0.0	2.3	0.0	0.13	0.00	0.11	0.00

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MAXIMUM POLE DEFORMATIONS CALCULATED(w.r.t. wind direction)

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MAST ELEV ft	.......DEFLECTIONS (ft)......... ..... HORIZONTAL ...... ALONG	ACROSS	DOWN	.........ROTATIONS (deg)......... .... TILT .... ALONG	ACROSS	TWIST
179.0	2.61A	-0.05K	0.05A	1.36A	-0.02K	0.00I
174.1	2.49A	-0.04K	0.05A	1.36A	-0.02K	0.00I
169.1	2.38A	-0.04K	0.05A	1.36A	-0.02K	0.00I
164.2	2.26A	-0.04K	0.04A	1.35A	-0.02K	0.00I
159.3	2.14A	-0.04K	0.04A	1.35A	-0.02K	0.00I
154.4	2.03A	-0.04K	0.04A	1.34A	-0.02K	0.00I
149.4	1.91A	-0.03K	0.03A	1.33A	-0.02K	0.00I
144.5	1.80A	-0.03K	0.03A	1.31A	-0.02K	0.00I
138.2	1.66A	-0.03K	0.03A	1.28A	-0.02K	0.00I
132.5	1.53A	-0.03K	0.03A	1.25A	-0.02K	0.00I
126.8	1.41A	-0.03K	0.02A	1.21A	-0.02K	0.00I
121.1	1.29A	-0.02K	0.02A	1.17A	-0.02K	0.00I
115.4	1.18A	-0.02K	0.02A	1.12A	-0.02K	0.00I
109.7	1.07A	-0.02K	0.02A	1.08A	-0.02K	0.00I
104.0	0.96A	-0.02K	0.01A	1.03A	-0.02K	0.00I
98.2	0.86A	-0.02K	0.01A	0.97A	-0.02K	0.00I
91.0	0.74A	-0.01K	0.01A	0.91A	-0.02K	0.00I
85.6	0.66A	-0.01K	0.01A	0.86A	-0.02K	0.00I
80.2	0.58A	-0.01K	0.01A	0.81A	-0.01K	0.00I
74.8	0.51A	-0.01K	0.01A	0.76A	-0.01K	0.00I
69.4	0.44A	-0.01K	0.01A	0.71A	-0.01K	0.00I
64.0	0.37A	-0.01K	0.00A	0.66A	-0.01K	0.00I
58.6	0.31A	-0.01K	0.00A	0.61A	-0.01K	0.00I
53.2	0.26A	0.00K	0.00A	0.55A	-0.01K	0.00I
44.7	0.18A	0.00K	0.00A	0.47A	-0.01K	0.00I
39.2	0.14A	0.00K	0.00A	0.41A	-0.01K	0.00I
33.6	0.10A	0.00K	0.00A	0.35A	-0.01K	0.00I
28.0	0.07A	0.00K	0.00A	0.29A	0.00K	0.00I
22.4	0.05A	0.00K	0.00A	0.23A	0.00K	0.00I
16.8	0.03A	0.00K	0.00A	0.18A	0.00K	0.00I
11.2	0.01A	0.00K	0.00A	0.12A	0.00K	0.00I
5.6	0.00A	0.00K	0.00G	0.06A	0.00K	0.00I

0.0	0.00A	0.00A	0.00A	0.00A	0.00A	0.00A

MAXIMUM ANTENNA AND REFLECTOR ROTATIONS

ELEV ft	ANT AZI deg	ANT TYPE	.... BEAM DEFLECTIONS (deg) .... ROLL	YAW	PITCH	TOTAL
165.0	0.0	HP	-1.329 J	0.014 K	1.355 A	1.355 A

MAXIMUM POLE FORCES CALCULATED(w.r.t. to wind direction)

MAST ELEV ft	TOTAL AXIAL kip	SHEAR.w.r.t.WIND.DIR ALONG kip	ACROSS kip	MOMENT.w.r.t.WIND.DIR ALONG ft-kip	ACROSS ft-kip	TORSION ft-kip
179.0						
	0.01 A	-0.06 I	0.04 K	0.11 A	0.09 K	0.00 K
	3.70 A	1.87 A	0.04 K	-2.05 B	-0.09 K	0.00 K
174.1						
	3.71 H	1.88 D	0.07 H	-2.11 J	-0.14 B	0.00 C
	4.33 H	2.00 D	0.07 H	-11.99 A	-0.30 E	0.01 H
169.1						
	4.33 H	2.07 H	-0.07 B	-11.91 D	-0.30 H	0.01 H
	7.62 H	3.76 A	-0.12 K	-24.30 H	0.54 B	0.18 D
164.2						
	7.62 H	3.83 A	-0.11 K	-24.22 H	0.54 B	0.18 D
	8.27 H	3.96 A	-0.11 K	-43.92 A	0.77 K	0.18 D
159.3						
	8.27 H	3.96 A	-0.16 K	-44.07 A	0.70 K	0.18 D
	8.95 H	4.10 A	-0.16 K	-64.89 A	1.51 K	0.18 D
154.4						
	8.95 H	4.06 A	-0.14 K	-64.78 A	1.44 K	0.18 D
	18.59 H	7.13 A	-0.14 K	-100.60 A	2.16 K	0.18 D
149.4						
	18.59 H	7.08 L	-0.13 L	-100.57 A	2.18 K	0.18 D
	19.29 H	7.22 L	-0.13 L	-137.51 A	2.67 K	0.18 D
144.5						
	19.28 H	7.19 L	-0.10 C	-137.59 A	2.69 K	0.18 D
	21.08 H	7.36 L	-0.10 C	-185.70 A	3.25 K	0.18 D
138.2						
	21.08 H	7.35 A	-0.13 K	-185.72 A	3.28 K	0.18 D
	21.94 H	7.51 A	-0.13 K	-230.88 A	4.04 K	0.18 D
132.5						
	21.94 H	7.55 A	-0.14 K	-230.92 A	4.07 K	0.18 D
	22.81 H	7.71 A	-0.14 K	-277.19 A	4.93 K	-0.18 I
126.8						
	22.81 H	7.72 A	-0.19 K	-277.24 A	4.91 K	-0.18 I
	23.71 H	7.89 A	-0.19 K	-324.59 A	6.03 K	-0.19 I
121.1						
	23.71 H	7.94 A	-0.18 K	-324.53 A	6.02 K	-0.19 I
	24.70 H	8.15 A	-0.18 K	-373.09 A	7.11 K	-0.20 I
115.4						
	24.70 H	8.11 A	-0.18 K	-373.06 A	7.09 K	-0.19 I
	27.83 H	9.51 A	-0.18 K	-428.13 A	8.15 K	-0.20 I
109.7						
	27.83 H	9.50 A	-0.18 E	-428.15 A	8.13 K	-0.20 I
	29.58 H	9.84 A	-0.18 E	-486.87 A	9.07 K	-0.21 I
104.0						

	29.58 H	9.86 A	-0.20 K	-486.86 A	9.07 K	-0.21 I
	30.62 H	10.07 A	-0.20 K	-546.78 A	10.23 K	-0.22 I
98.2						
	30.62 H	10.12 A	-0.22 K	-546.78 A	10.24 K	-0.22 I
	34.08 H	10.47 A	-0.22 K	-625.18 A	11.89 K	-0.23 I
91.0						
	34.08 H	10.46 A	-0.21 K	-625.09 A	11.93 K	-0.23 I
	35.23 H	10.62 A	-0.21 K	-684.86 A	13.11 K	-0.24 I
85.6						
	35.23 H	10.51 A	-0.22 K	-684.83 A	13.16 K	-0.24 I
	36.40 H	10.68 A	-0.22 K	-744.85 A	14.40 K	-0.26 I
80.2						
	36.40 H	10.69 A	-0.19 K	-744.98 A	14.41 K	-0.26 I
	37.60 H	10.86 A	-0.19 K	-805.77 A	15.46 K	-0.27 I
74.8						
	37.60 H	10.85 G	-0.17 C	-805.92 A	15.46 K	-0.27 I
	38.79 H	11.00 G	-0.17 C	-867.41 A	16.35 K	-0.28 I
69.4						
	38.79 H	11.09 A	-0.17 K	-867.46 A	16.32 K	-0.28 I
	40.02 H	11.26 A	-0.17 K	-930.19 A	17.29 K	-0.28 I
64.0						
	40.02 H	11.27 A	-0.18 E	-930.22 A	17.32 K	-0.28 I
	41.26 H	11.43 A	-0.18 E	-993.86 A	18.32 K	-0.29 I
58.6						
	41.26 H	11.44 A	-0.17 K	-993.86 A	18.34 K	-0.29 I
	42.54 H	11.61 A	-0.17 K	-1058.29 A	19.31 K	-0.30 I
53.2						
	42.54 H	11.61 A	-0.18 K	-1058.30 A	19.32 K	-0.30 I
	46.62 H	11.87 A	-0.18 K	-1161.48 A	20.92 K	-0.31 I
44.7						
	46.62 H	11.89 A	-0.18 K	-1161.49 A	20.94 K	-0.31 I
	47.99 H	12.05 A	-0.18 K	-1230.49 A	21.94 K	-0.31 I
39.2						
	47.99 H	12.06 A	-0.19 K	-1230.48 A	21.94 K	-0.31 I
	49.39 H	12.23 A	-0.19 K	-1300.21 A	22.99 K	-0.32 I
33.6						
	49.39 H	12.18 G	-0.17 K	-1300.22 A	22.99 K	-0.32 I
	50.80 H	12.34 G	-0.17 K	-1370.22 A	23.96 K	-0.33 I
28.0						
	50.80 H	12.35 G	-0.17 K	-1370.22 A	23.97 K	-0.33 I
	52.25 H	12.51 G	-0.17 K	-1440.90 A	24.94 K	-0.33 I
22.4						
	52.25 H	12.49 G	-0.17 K	-1440.90 A	24.94 K	-0.33 I
	53.70 H	12.64 G	-0.17 K	-1512.19 A	25.92 K	-0.33 I
16.8						
	53.70 H	12.63 G	-0.17 K	-1512.19 A	25.92 K	-0.33 I
	55.18 H	12.78 G	-0.17 K	-1584.00 A	26.90 K	-0.34 I
11.2						
	55.18 H	12.78 G	-0.18 K	-1584.00 A	26.90 K	-0.34 I
	56.70 H	12.93 G	-0.18 K	-1656.35 A	27.90 K	-0.34 I
5.6						
	56.70 H	12.93 G	-0.18 K	-1656.35 A	27.90 K	-0.34 I
	58.22 H	13.08 G	-0.18 K	-1729.19 A	28.88 K	-0.34 I
base reaction	58.22 H	-13.08 G	0.18 K	1729.19 A	-28.88 K	0.34 I

COMPLIANCE WITH 4.8.2 & 4.5.4

ELEV	AXIAL	BENDING	SHEAR +	TOTAL	SATISFIED	D/t(w/t)	MAX

			TORSIONAL				ALLOWED
ft							
179.00							
	0.00A	0.00A	0.00I	0.00A	YES	19.26A	45.2
	0.00A	0.00B	0.00A	0.00B	YES	19.79A	45.2
174.07							
	0.00H	0.00J	0.00D	0.00J	YES	19.79A	45.2
	0.00H	0.01A	0.00D	0.01A	YES	20.33A	45.2
169.14							
	0.00H	0.01D	0.00H	0.01D	YES	20.33A	45.2
	0.00H	0.01H	0.00A	0.01H	YES	20.86A	45.2
164.21							
	0.00H	0.01H	0.00A	0.01H	YES	20.86A	45.2
	0.00H	0.02A	0.00A	0.02A	YES	21.40A	45.2
159.29							
	0.00H	0.02A	0.00A	0.02A	YES	21.40A	45.2
	0.00H	0.03A	0.00A	0.03A	YES	21.93A	45.2
154.36							
	0.00H	0.03A	0.00A	0.03A	YES	21.93A	45.2
	0.01H	0.04A	0.01A	0.05A	YES	22.47A	45.2
149.43							
	0.01H	0.04A	0.01L	0.05A	YES	22.47A	45.2
	0.01H	0.06A	0.01L	0.06A	YES	23.00A	45.2
144.50							
	0.01H	0.06A	0.01L	0.06A	YES	23.00A	45.2
	0.01H	0.07A	0.01L	0.08A	YES	23.68A	45.2
138.25							
	0.01H	0.07A	0.01A	0.08A	YES	23.33A	45.2
	0.01H	0.09A	0.01A	0.10A	YES	23.95A	45.2
132.54							
	0.01H	0.09A	0.01A	0.10A	YES	23.95A	45.2
	0.01H	0.10A	0.01A	0.11A	YES	24.57A	45.2
126.82							
	0.01H	0.10A	0.01A	0.11A	YES	24.57A	45.2
	0.01H	0.12A	0.01A	0.12A	YES	25.19A	45.2
121.11							
	0.01H	0.12A	0.01A	0.12A	YES	25.19A	45.2
	0.01H	0.13A	0.01A	0.14A	YES	25.81A	45.2
115.39							
	0.01H	0.13A	0.01A	0.14A	YES	25.81A	45.2
	0.01H	0.14A	0.01A	0.15A	YES	26.43A	45.2
109.68							
	0.01H	0.14A	0.01A	0.15A	YES	26.43A	45.2
	0.01H	0.16A	0.01A	0.16A	YES	27.05A	45.2
103.96							
	0.01H	0.16A	0.01A	0.16A	YES	27.05A	45.2
	0.01H	0.17A	0.01A	0.18A	YES	27.67A	45.2
98.25							
	0.01H	0.13A	0.01A	0.14A	YES	23.00A	45.2
	0.01H	0.14A	0.01A	0.15A	YES	23.66A	45.2
91.00							
	0.01H	0.15A	0.01A	0.15A	YES	23.37A	45.2
	0.01H	0.15A	0.01A	0.16A	YES	23.85A	45.2
85.61							
	0.01H	0.15A	0.01A	0.16A	YES	23.85A	45.2
	0.01H	0.16A	0.01A	0.17A	YES	24.34A	45.2
80.21							
	0.01H	0.16A	0.01A	0.17A	YES	24.34A	45.2
	0.01H	0.17A	0.01A	0.18A	YES	24.83A	45.2
74.82							
	0.01H	0.17A	0.01G	0.18A	YES	24.83A	45.2

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         0.01H    0.18A    0.01G    0.19A    YES     25.32A      45.2
 69.43 .....................................................................
         0.01H    0.18A    0.01A    0.19A    YES     25.32A      45.2

         0.01H    0.19A    0.01A    0.20A    YES     25.81A      45.2
 64.04 .....................................................................
         0.01H    0.19A    0.01A    0.20A    YES     25.81A      45.2

         0.01H    0.19A    0.01A    0.20A    YES     26.29A      45.2
 58.64 .....................................................................
         0.01H    0.19A    0.01A    0.20A    YES     26.29A      45.2

         0.01H    0.20A    0.01A    0.21A    YES     26.78A      45.2
 53.25 .....................................................................
         0.01H    0.20A    0.01A    0.21A    YES     26.78A      45.2

         0.01H    0.21A    0.01A    0.22A    YES     27.55A      45.2
 44.75 .....................................................................
         0.01H    0.21A    0.01A    0.22A    YES     27.20A      45.2

         0.01H    0.22A    0.01A    0.23A    YES     27.71A      45.2
 39.16 .....................................................................
         0.01H    0.22A    0.01A    0.23A    YES     27.71A      45.2

         0.01H    0.23A    0.01A    0.24A    YES     28.21A      45.2
 33.56 .....................................................................
         0.01H    0.23A    0.01G    0.24A    YES     28.21A      45.2

         0.01H    0.23A    0.01G    0.24A    YES     28.72A      45.2
 27.97 .....................................................................
         0.01H    0.23A    0.01G    0.24A    YES     28.72A      45.2

         0.01H    0.24A    0.01G    0.25A    YES     29.22A      45.2
 22.37 .....................................................................
         0.01H    0.24A    0.01G    0.25A    YES     29.22A      45.2

         0.01H    0.25A    0.01G    0.26A    YES     29.73A      45.2
 16.78 .....................................................................
         0.01H    0.25A    0.01G    0.26A    YES     29.73A      45.2

         0.01H    0.25A    0.01G    0.26A    YES     30.24A      45.2
 11.19 .....................................................................
         0.01H    0.25A    0.01G    0.26A    YES     30.24A      45.2

         0.01H    0.26A    0.01G    0.27A    YES     30.74A      45.2
  5.59 .....................................................................
         0.01H    0.26A    0.01G    0.27A    YES     30.74A      45.2

         0.01H    0.26A    0.01G    0.27A    YES     31.25A      45.2
  0.00 .....................................................................

MAXIMUM LOADS ONTO FOUNDATION(w.r.t. wind direction)
====================================================

    DOWN    SHEAR.w.r.t.WIND.DIR   MOMENT.w.r.t.WIND.DIR     TORSION
               ALONG      ACROSS       ALONG      ACROSS
     kip         kip         kip      ft-kip      ft-kip      ft-kip

   58.22       13.08       -0.18    -1729.19       28.88       -0.34
       H           G           K           A           K           I


===============================================================================
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Seismic Load Effects
Equivalent Lateral Force Procedure
ANSI/TIA-222-H

Parameters

Risk Category	II
R	1.500
S_S	0.142
S_1	0.051
Site Class	D (default)
T_L (sec)	6.000
F_a	1.600
F_v	2.400
S_{MS}	0.227
S_{M1}	0.122
S_{DS}	0.151
S_{D1}	0.082
T_s	0.543
I_e	1.000
Ω	1.500
C_S	0.030
E (ksi)	29,000
I_{top} (in^4)	5,709
I_{bot} (in^4)	47,569
I_{avg} (in^4)	26,639
g (in/s^2)	386.4
W_t (kips)	58.278
W_u (kips)	11.413
W_L (kips)	46.864
L_p (in)	2148
f_1 (Hertz)	0.319
T (sec)	3.133
k_e	2.0000
V_s (kips)	1.748
Seismic Design Category	B

Vertical Distribution of Seismic Forces

Description	h_i (ft.)	w_i (kips)	W_u (kips)	$w_i h_i^{ke}$	F_{sz} or E_h (kips)	E_V (kips)	1.2 D + 1.0 E_V (kips)	0.9 D - 1.0 E_V (kips)
Line Deadload	175.00	1.0605	0.0000	32,477.8125	0.0751	0.0320	1.3046	0.9224
Mount/Antenna Load	175.00	2.0098	2.0098	61,550.1250	0.1423	0.0607	2.4725	1.7481
Step Bolts/Safety Climb Load	174.50	0.0126	0.0000	383.6732	0.0009	0.0004	0.0155	0.0109
Line Deadload	165.00	1.4091	0.0000	38,362.7475	0.0887	0.0426	1.7335	1.2256
Mount/Antenna Load	165.00	0.2950	0.2950	8,031.3750	0.0186	0.0089	0.3629	0.2566
Mount/Antenna Load	165.00	1.1083	1.1083	30,173.4675	0.0697	0.0335	1.3635	0.9640
Step Bolts/Safety Climb Load	165.00	0.0140	0.0000	381.1500	0.0009	0.0004	0.0172	0.0122
Structure - Section 1	158.62	5.4341	0.0000	136,723.6101	0.3160	0.1641	6.6850	4.7266
Step Bolts/Safety Climb Load	155.00	0.0140	0.0000	336.3500	0.0008	0.0004	0.0172	0.0122
Antenna Load	154.00	8.0000	8.0000	189,728.0000	0.4386	0.2416	9.8416	6.9584
Line Deadload	154.00	0.9610	0.0000	22,791.0760	0.0527	0.0290	1.1822	0.8359
Step Bolts/Safety Climb Load	145.00	0.0140	0.0000	294.3500	0.0007	0.0004	0.0172	0.0122
Step Bolts/Safety Climb Load	135.00	0.0140	0.0000	255.1500	0.0006	0.0004	0.0172	0.0122
Step Bolts/Safety Climb Load	125.00	0.0140	0.0000	218.7500	0.0005	0.0004	0.0172	0.0122
Structure - Section 2	117.75	8.4390	0.0000	117,007.2624	0.2705	0.2549	10.3817	7.3402
Antenna Load	115.72	0.0285	0.0000	381.6469	0.0009	0.0009	0.0351	0.0248
Line Deadload	115.72	0.0625	0.0000	836.9449	0.0019	0.0019	0.0769	0.0544
Step Bolts/Safety Climb Load	115.00	0.0140	0.0000	185.1500	0.0004	0.0004	0.0172	0.0122
Line Deadload	114.00	0.6156	0.0000	8,000.3376	0.0185	0.0186	0.7573	0.5354
Mount/Antenna Load	114.00	1.5808	0.0000	20,544.0768	0.0475	0.0477	1.9447	1.3750
Antenna Load	109.00	0.0500	0.0000	594.0500	0.0014	0.0015	0.0615	0.0435
Line Deadload	109.00	0.0436	0.0000	518.0116	0.0012	0.0013	0.0536	0.0379
Mount Load	109.00	0.7073	0.0000	8,403.4313	0.0194	0.0214	0.8702	0.6152
Step Bolts/Safety Climb Load	105.00	0.0140	0.0000	154.3500	0.0004	0.0004	0.0172	0.0122
Antenna Load	100.72	0.0285	0.0000	289.1188	0.0007	0.0009	0.0351	0.0248
Line Deadload	100.72	0.0544	0.0000	551.8618	0.0013	0.0016	0.0669	0.0474
Step Bolts/Safety Climb Load	95.00	0.0140	0.0000	126.3500	0.0003	0.0004	0.0172	0.0122
Mount Load	94.00	0.7073	0.0000	6,249.7028	0.0144	0.0214	0.8702	0.6152
Step Bolts/Safety Climb Load	85.00	0.0140	0.0000	101.1500	0.0002	0.0004	0.0172	0.0122
Step Bolts/Safety Climb Load	75.00	0.0140	0.0000	78.7500	0.0002	0.0004	0.0172	0.0122
Structure - Section 3	71.50	11.9021	0.0000	60,846.5107	0.1407	0.3594	14.6419	10.3525
Step Bolts/Safety Climb Load	65.00	0.0140	0.0000	59.1500	0.0001	0.0004	0.0172	0.0122
Step Bolts/Safety Climb Load	55.00	0.0140	0.0000	42.3500	0.0001	0.0004	0.0172	0.0122
Step Bolts/Safety Climb Load	45.00	0.0140	0.0000	28.3500	0.0001	0.0004	0.0172	0.0122
Step Bolts/Safety Climb Load	35.00	0.0140	0.0000	17.1500	0.0000	0.0004	0.0172	0.0122
Structure - Section 4	26.62	13.5435	0.0000	9,597.2546	0.0222	0.4090	16.6612	11.7801
Step Bolts/Safety Climb Load	25.00	0.0140	0.0000	8.7500	0.0000	0.0004	0.0172	0.0122

Seismic Load Effects
Equivalent Lateral Force Procedure
ANSI/TIA-222-H

Vertical Distribution of Seismic Forces

Description	h_i (ft.)	w_i (kips)	W_u (kips)	$w_i h_i^{ke}$	F_{sz} or E_h (kips)	E_V (kips)	1.2 D + 1.0 E_V (kips)	0.9 D - 1.0 E_V (kips)
Step Bolts/Safety Climb Load	15.00	0.0140	0.0000	3.1500	0.0000	0.0004	0.0172	0.0122
	Σ	58.28	11.4131	756,332.50	1.75	1.76	71.69	50.69

Sabre Industries™
INNOVATION DELIVERED

SO#:	25-3277-JDS-R1-RB
Site Name:	US-NY-2029, NY
Date:	7/15/2025

Round Base Plate and Anchor Rods, per ANSI/TIA 222-H

Pole Data

Diameter:	68.460	in (flat to flat)
Thickness:	0.375	in
Yield (Fy):	65	ksi
# of Sides:	18	"0" IF Round
Strength (Fu):	80	ksi

Reactions

Moment, Mu:	6352.75	ft-kips
Axial, Pu:	69.92	kips
Shear, Vu:	48.07	kips

Anchor Rod Data

Quantity:	18		
Diameter:	2.25	in	
Rod Material:	A615		
Strength (Fu):	100	ksi	
Yield (Fy):	75	ksi	
BC Diam. (in):	75.5	BC Override:	

Plate Data

Diameter (in):	81.25	Dia. Override:	
Thickness:	2	in	
Yield (Fy):	50	ksi	
Eff Width/Rod:	12.07	in	
Drain Hole:	2.625	in. diameter	
Drain Location:	32.25	in. center of pole to center of drain hole	
Center Hole:	56.5	in. diameter	

Anchor Rod Results

(per 4.9.9)

Maximum Put:	221.47	Kips
Φt*Rnt:	243.75	Kips
Vu:	2.67	Kips
Φv*Rnv:	149.10	Kips
Tension Interaction Ratio:	0.83	
Maximum Puc:	228.26	Kips
Φc*Rnc:	268.39	Kips
Vu:	2.67	Kips
Φc*Rnvc:	120.77	Kips
Compression Interaction Ratio:	0.85	
Maximum Interaction Ratio:	**85.1%**	**Pass**

Base Plate Results

Base Plate (Mu/Z):	42.0	ksi	
Allowable Φ*Fy:	45.0	ksi	(per AISC)
Base Plate Interaction Ratio:	**93.4%**	**Pass**	

MAT FOUNDATION DESIGN BY SABRE INDUSTRIES
180' Monopole PTI SERVICES US-NY-2029, NY (25-3277-JDS-R1) 07/15/25 TTW

Overall Loads:			
Factored Moment (ft-kips)	6428.91		
Factored Axial (kips)	70.19		
Factored Shear (kips)	48.56		
Bearing Design Strength (ksf)	2.985	Max. Net Bearing Press. (ksf)	2.97
Water Table Below Grade (ft)	**27.2**		
Width of Mat (ft)	29	Allowable Bearing Pressure (ksf)	**1.99**
Thickness of Mat (ft)	1.75	Safety Factor	**2.00**
Depth to Bottom of Slab (ft)	6	Ultimate Bearing Pressure (ksf)	3.98
Quantity of Bolts in Bolt Circle	18	Bearing Φs	0.75
Bolt Circle Diameter (in)	75.5		
Effective Anchor Bolt Embedment (in)	66.5		
Diameter of Pier (ft)	8	Minimum Pier Diameter (ft)	8.00
Ht. of Pier Above Ground (ft)	0.5	Equivalent Square b (ft)	7.09
Ht. of Pier Below Ground (ft)	4.25	Square Pier? (Y/N)	**N**
Quantity of Bars in Mat	51		
Bar Diameter in Mat (in)	1.27		
Area of Bars in Mat (in^2)	64.61		
Spacing of Bars in Mat (in)	6.81	Recommended Spacing (in)	5 to 12
Quantity of Bars Pier	50		
Bar Diameter in Pier (in)	1		
Tie Bar Diameter in Pier (in)	0.625		
Spacing of Ties (in)	4		
Area of Bars in Pier (in^2)	39.27	Minimum Pier A_s (in^2)	36.19
Spacing of Bars in Pier (in)	5.51	Recommended Spacing (in)	5 to 12
f'c (ksi)	4.5		
fy (ksi)	60		
Unit Wt. of Soil (kcf)	**0.117**		
Unit Wt. of Concrete (kcf)	0.15		
Volume of Concrete (yd^3)	63.35		
Two-Way Shear Action:			
Average d (in)	16.73		
ϕv_c (ksi)	0.183	v_u (ksi)	0.148
$\phi v_c = \phi(2 + 4/\beta_c)f'_c{}^{1/2}$	0.302		
$\phi v_c = \phi(\alpha_s d/b_o + 2)f'_c{}^{1/2}$	0.183	J (in^3)	1.185E+07
$\phi v_c = \phi 4 f'_c{}^{1/2}$	0.201	c + d (in)	101.81
Shear perimeter, b_o (in)	407.23	$0.40M_{sc}$ (ft-kips)	2663.8
β_c	1		
One-Way Shear:			
ϕV_c (kips)	585.8	V_u (kips)	412.8
Stability:			
Overturning Design Strength (ft-k)	9242.8	Total Applied M (ft-k)	6744.6

Pier-Slab Transfer by Flexure:

b_{slab} (ft)	13.25		
ØM$_n$ (ft-kips)	4057.7	0.60M$_{sc}$ (ft-kips)	3995.7

Pier Design:

ϕV_n (kips)	1275.6	V_u (kips)	48.6
$\phi V_c=\phi 2(1+N_u/(2000A_g))f'_c{}^{1/2}b_wd$	745.5		
V_s (kips)	706.9	*** V_s max = 4 $f'_c{}^{1/2}b_wd$ (kips)	1978.3
Maximum Spacing (in)	7.62	(Only if Shear Ties are Required)	
Actual Hook Development (in)	15.46	Req'd Hook Development l_{dh} (in) - Tension	12.52
		Req'd Hook Development l_{dc} (in) - Compression	13.50

Flexure in Slab:

ϕM_n (ft-kips)	4440.5	M_u (ft-kips)	2996.2
a (in)	2.91		
Steel Ratio	0.01110		
β_1	0.825		
Maximum Steel Ratio (ρ_t)	0.0197		
Minimum Steel Ratio	0.0018		
Rebar Development in Pad (in)	123.00	Required Development in Pad (in)	34.08

Condition	1 is OK, 0 Fails
Maximum Soil Bearing Pressure	1
Pier Area of Steel	1
Pier Shear	1
Interaction Diagram	1
Two-Way Shear Action	1
One-Way Shear Action	1
Overturning	1
Flexure	1
Steel Ratio	1
Length of Development in Pad	1
Hook Development	1
Anchor Bolt Pullout	1
Anchor Bolt Punching Shear	1

```
================================================================================
                    LPile for Windows, Version 2019-11.009

                Analysis of Individual Piles and Drilled Shafts
               Subjected to Lateral Loading Using the p-y Method
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================================================================================

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--------------------------------------------------------------------------------
                            Files Used for Analysis
--------------------------------------------------------------------------------

Path to file locations:
\Program Files (x86)\Ensoft\Lpile2019\files\

Name of input data file:
25-3277-JDS-R1.lp11d

Name of output report file:
25-3277-JDS-R1.lp11o

Name of plot output file:
25-3277-JDS-R1.lp11p

Name of runtime message file:
25-3277-JDS-R1.lp11r


--------------------------------------------------------------------------------
                           Date and Time of Analysis
--------------------------------------------------------------------------------

               Date:  April 17, 2025             Time:   9:16:14


--------------------------------------------------------------------------------
                                Problem Title
--------------------------------------------------------------------------------

Site         : US-NY-2029, NY


Tower        : 180' Monopole


Prepared for : PTI SERVICES


Job Number   : 25-3277-JDS-R1


Engineer     : TTW



--------------------------------------------------------------------------------
                          Program Options and Settings
--------------------------------------------------------------------------------

Computational Options:
```

```
 - Conventional Analysis
Engineering Units Used for Data Input and Computations:
 - US Customary System Units (pounds, feet, inches)

Analysis Control Options:
 - Maximum number of iterations allowed                =          999
 - Deflection tolerance for convergence                =    1.0000E-05 in
 - Maximum allowable deflection                        =      100.0000 in
 - Number of pile increments                           =          100

Loading Type and Number of Cycles of Loading:
 - Static loading specified

 - Use of p-y modification factors for p-y curves not selected
 - Analysis uses layering correction (Method of Georgiadis)
 - No distributed lateral loads are entered
 - Loading by lateral soil movements acting on pile not selected
 - Input of shear resistance at the pile tip not selected
 - Input of moment resistance at the pile tip not selected
 - Input of side resistance moment along pile not selected
 - Computation of pile-head foundation stiffness matrix not selected
 - Push-over analysis of pile not selected
 - Buckling analysis of pile not selected

Output Options:
 - Output files use decimal points to denote decimal symbols.
 - Report only summary tables of pile-head deflection, maximum bending moment,
   and maximum shear force in output report file.
 - No p-y curves to be computed and reported for user-specified depths
 - Print using wide report formats



--------------------------------------------------------------------------------
                     Pile Structural Properties and Geometry
--------------------------------------------------------------------------------

Number of pile sections defined                        =            1
Total length of pile                                   =       27.500 ft
Depth of ground surface below top of pile              =       0.5000 ft

Pile diameters used for p-y curve computations are defined using 2 points.

p-y curves are computed using pile diameter values interpolated with depth over
the length of the pile. A summary of values of pile diameter vs. depth follows.

            Depth Below           Pile
Point        Pile Head          Diameter
 No.           feet              inches
-----      -------------     -------------
  1            0.000            96.0000
  2           27.500            96.0000


Input Structural Properties for Pile Sections:
----------------------------------------------

Pile Section No. 1:

   Section 1 is a round drilled shaft, bored pile, or CIDH pile
   Length of section                                   =     27.500000 ft
   Shaft Diameter                                      =     96.000000 in
   Shear capacity of section                           =        0.0000 lbs



--------------------------------------------------------------------------------
                       Ground Slope and Pile Batter Angles
--------------------------------------------------------------------------------

Ground Slope Angle                                     =        0.000 degrees
                                                       =        0.000 radians

Pile Batter Angle                                      =        0.000 degrees
                                                       =        0.000 radians


--------------------------------------------------------------------------------
                       Soil and Rock Layering Information
```

```
--------------------------------------------------------------------------------

The soil profile is modelled using 10 layers

Layer 1 is sand, p-y criteria by Reese et al., 1974

   Distance from top of pile to top of layer          =     0.500000 ft
   Distance from top of pile to bottom of layer       =     2.500000 ft
   Effective unit weight at top of layer              =   115.000000 pcf
   Effective unit weight at bottom of layer           =   115.000000 pcf
   Friction angle at top of layer                     =    30.000000 deg.
   Friction angle at bottom of layer                  =    30.000000 deg.
   Subgrade k at top of layer                         =    90.000000 pci
   Subgrade k at bottom of layer                      =    90.000000 pci


Layer 2 is sand, p-y criteria by Reese et al., 1974

   Distance from top of pile to top of layer          =     2.500000 ft
   Distance from top of pile to bottom of layer       =     4.500000 ft
   Effective unit weight at top of layer              =   120.000000 pcf
   Effective unit weight at bottom of layer           =   120.000000 pcf
   Friction angle at top of layer                     =    36.000000 deg.
   Friction angle at bottom of layer                  =    36.000000 deg.
   Subgrade k at top of layer                         =   225.000000 pci
   Subgrade k at bottom of layer                      =   225.000000 pci


Layer 3 is sand, p-y criteria by Reese et al., 1974

   Distance from top of pile to top of layer          =     4.500000 ft
   Distance from top of pile to bottom of layer       =     6.500000 ft
   Effective unit weight at top of layer              =   120.000000 pcf
   Effective unit weight at bottom of layer           =   120.000000 pcf
   Friction angle at top of layer                     =    35.000000 deg.
   Friction angle at bottom of layer                  =    35.000000 deg.
   Subgrade k at top of layer                         =    90.000000 pci
   Subgrade k at bottom of layer                      =    90.000000 pci


Layer 4 is sand, p-y criteria by Reese et al., 1974

   Distance from top of pile to top of layer          =     6.500000 ft
   Distance from top of pile to bottom of layer       =     8.500000 ft
   Effective unit weight at top of layer              =   115.000000 pcf
   Effective unit weight at bottom of layer           =   115.000000 pcf
   Friction angle at top of layer                     =    30.000000 deg.
   Friction angle at bottom of layer                  =    30.000000 deg.
   Subgrade k at top of layer                         =    90.000000 pci
   Subgrade k at bottom of layer                      =    90.000000 pci


Layer 5 is sand, p-y criteria by Reese et al., 1974

   Distance from top of pile to top of layer          =     8.500000 ft
   Distance from top of pile to bottom of layer       =    13.500000 ft
   Effective unit weight at top of layer              =   115.000000 pcf
   Effective unit weight at bottom of layer           =   115.000000 pcf
   Friction angle at top of layer                     =    31.000000 deg.
   Friction angle at bottom of layer                  =    31.000000 deg.
   Subgrade k at top of layer                         =    90.000000 pci
   Subgrade k at bottom of layer                      =    90.000000 pci


Layer 6 is stiff clay without free water

   Distance from top of pile to top of layer          =    13.500000 ft
   Distance from top of pile to bottom of layer       =    18.500000 ft
   Effective unit weight at top of layer              =   110.000000 pcf
   Effective unit weight at bottom of layer           =   110.000000 pcf
   Undrained cohesion at top of layer                 =   500.000000 psf
   Undrained cohesion at bottom of layer              =   500.000000 psf
   Epsilon-50 at top of layer                         =     0.020000
   Epsilon-50 at bottom of layer                      =     0.020000


Layer 7 is soft clay, p-y criteria by Matlock, 1970

   Distance from top of pile to top of layer          =    18.500000 ft
   Distance from top of pile to bottom of layer       =    23.500000 ft
   Effective unit weight at top of layer              =   110.000000 pcf
```

```
   Effective unit weight at bottom of layer           =   110.000000 pcf
   Undrained cohesion at top of layer                 =   400.000000 psf
   Undrained cohesion at bottom of layer              =   400.000000 psf
   Epsilon-50 at top of layer                         =     0.020000
   Epsilon-50 at bottom of layer                      =     0.020000
```

Layer 8 is sand, p-y criteria by Reese et al., 1974

```
   Distance from top of pile to top of layer          =    23.500000 ft
   Distance from top of pile to bottom of layer       =    25.500000 ft
   Effective unit weight at top of layer              =   115.000000 pcf
   Effective unit weight at bottom of layer           =   115.000000 pcf
   Friction angle at top of layer                     =    30.000000 deg.
   Friction angle at bottom of layer                  =    30.000000 deg.
   Subgrade k at top of layer                         =    90.000000 pci
   Subgrade k at bottom of layer                      =    90.000000 pci
```

Layer 9 is stiff clay without free water

```
   Distance from top of pile to top of layer          =    25.500000 ft
   Distance from top of pile to bottom of layer       =    27.500000 ft
   Effective unit weight at top of layer              =   130.000000 pcf
   Effective unit weight at bottom of layer           =   130.000000 pcf
   Undrained cohesion at top of layer                 =        4000. psf
   Undrained cohesion at bottom of layer              =        4000. psf
   Epsilon-50 at top of layer                         =     0.005000
   Epsilon-50 at bottom of layer                      =     0.005000
```

Layer 10 is stiff clay without free water

```
   Distance from top of pile to top of layer          =    27.500000 ft
   Distance from top of pile to bottom of layer       =    38.500000 ft
   Effective unit weight at top of layer              =    67.600000 pcf
   Effective unit weight at bottom of layer           =    67.600000 pcf
   Undrained cohesion at top of layer                 =        4000. psf
   Undrained cohesion at bottom of layer              =        4000. psf
   Epsilon-50 at top of layer                         =     0.005000
   Epsilon-50 at bottom of layer                      =     0.005000
```

(Depth of the lowest soil layer extends 11.000 ft below the pile tip)

Summary of Input Soil Properties

Layer Num.	Soil Type Name (p-y Curve Type)	Layer Depth ft	Effective Unit Wt. pcf	Cohesion psf	Angle of Friction deg.	E50 or krm	kpy pci
1	Sand	0.5000	115.0000	--	30.0000	--	90.0000
	(Reese, et al.)	2.5000	115.0000	--	30.0000	--	90.0000
2	Sand	2.5000	120.0000	--	36.0000	--	225.0000
	(Reese, et al.)	4.5000	120.0000	--	36.0000	--	225.0000
3	Sand	4.5000	120.0000	--	35.0000	--	90.0000
	(Reese, et al.)	6.5000	120.0000	--	35.0000	--	90.0000
4	Sand	6.5000	115.0000	--	30.0000	--	90.0000
	(Reese, et al.)	8.5000	115.0000	--	30.0000	--	90.0000
5	Sand	8.5000	115.0000	--	31.0000	--	90.0000
	(Reese, et al.)	13.5000	115.0000	--	31.0000	--	90.0000
6	Stiff Clay	13.5000	110.0000	500.0000	--	0.02000	--

	w/o Free Water	18.5000	110.0000	500.0000	--	0.02000	--
7	Soft	18.5000	110.0000	400.0000	--	0.02000	--
	Clay	23.5000	110.0000	400.0000	--	0.02000	--
8	Sand	23.5000	115.0000	--	30.0000	--	90.0000
	(Reese, et al.)	25.5000	115.0000	--	30.0000	--	90.0000
9	Stiff Clay	25.5000	130.0000	4000.	--	0.00500	--
	w/o Free Water	27.5000	130.0000	4000.	--	0.00500	--
10	Stiff Clay	27.5000	67.6000	4000.	--	0.00500	--
	w/o Free Water	38.5000	67.6000	4000.	--	0.00500	--

Static Loading Type

Static loading criteria were used when computing p-y curves for all analyses.

Pile-head Loading and Pile-head Fixity Conditions

Number of loads specified = 2

Load No.	Load Type	Condition 1	Condition 2	Axial Thrust Force, lbs	Compute Top y vs. Pile Length	Run Analysis
1	1	V = 64747. lbs	M = 102862560. in-lbs	93587.	No	Yes
2	1	V = 13220. lbs	M = 20877480. in-lbs	58210.	No	Yes

V = shear force applied normal to pile axis
M = bending moment applied to pile head
y = lateral deflection normal to pile axis
S = pile slope relative to original pile batter angle
R = rotational stiffness applied to pile head
Values of top y vs. pile lengths can be computed only for load types with specified shear loading (Load Types 1, 2, and 3).
Thrust force is assumed to be acting axially for all pile batter angles.

Computations of Nominal Moment Capacity and Nonlinear Bending Stiffness

Axial thrust force values were determined from pile-head loading conditions

Number of Pile Sections Analyzed = 1

Pile Section No. 1:

Dimensions and Properties of Drilled Shaft (Bored Pile):

Length of Section	=	27.500000 ft
Shaft Diameter	=	96.000000 in
Concrete Cover Thickness (to edge of long. rebar)	=	3.625000 in
Number of Reinforcing Bars	=	52 bars
Yield Stress of Reinforcing Bars	=	60000. psi
Modulus of Elasticity of Reinforcing Bars	=	29000000. psi
Gross Area of Shaft	=	7238. sq. in.
Total Area of Reinforcing Steel	=	40.840704 sq. in.
Area Ratio of Steel Reinforcement	=	0.56 percent
Edge-to-Edge Bar Spacing	=	4.298213 in
Maximum Concrete Aggregate Size	=	0.750000 in
Ratio of Bar Spacing to Aggregate Size	=	5.73

Offset of Center of Rebar Cage from Center of Pile = 0.0000 in

Axial Structural Capacities:

Nom. Axial Structural Capacity = 0.85 Fc Ac + Fy As = 29980.454 kips
Tensile Load for Cracking of Concrete = -3305.543 kips
Nominal Axial Tensile Capacity = -2450.442 kips

Reinforcing Bar Dimensions and Positions Used in Computations:

Bar Number	Bar Diam. inches	Bar Area sq. in.	X inches	Y inches
1	1.000000	0.785398	43.875000	0.00000
2	1.000000	0.785398	43.555102	5.288547
3	1.000000	0.785398	42.600072	10.499975
4	1.000000	0.785398	41.023838	15.558289
5	1.000000	0.785398	38.849383	20.389729
6	1.000000	0.785398	36.108417	24.923841
7	1.000000	0.785398	32.840909	29.094507
8	1.000000	0.785398	29.094507	32.840909
9	1.000000	0.785398	24.923841	36.108417
10	1.000000	0.785398	20.389729	38.849383
11	1.000000	0.785398	15.558289	41.023838
12	1.000000	0.785398	10.499975	42.600072
13	1.000000	0.785398	5.288547	43.555102
14	1.000000	0.785398	0.00000	43.875000
15	1.000000	0.785398	-5.288547	43.555102
16	1.000000	0.785398	-10.499975	42.600072
17	1.000000	0.785398	-15.558289	41.023838
18	1.000000	0.785398	-20.389729	38.849383
19	1.000000	0.785398	-24.923841	36.108417
20	1.000000	0.785398	-29.094507	32.840909
21	1.000000	0.785398	-32.840909	29.094507
22	1.000000	0.785398	-36.108417	24.923841
23	1.000000	0.785398	-38.849383	20.389729
24	1.000000	0.785398	-41.023838	15.558289
25	1.000000	0.785398	-42.600072	10.499975
26	1.000000	0.785398	-43.555102	5.288547
27	1.000000	0.785398	-43.875000	0.00000
28	1.000000	0.785398	-43.555102	-5.288547
29	1.000000	0.785398	-42.600072	-10.499975
30	1.000000	0.785398	-41.023838	-15.558289
31	1.000000	0.785398	-38.849383	-20.389729
32	1.000000	0.785398	-36.108417	-24.923841
33	1.000000	0.785398	-32.840909	-29.094507
34	1.000000	0.785398	-29.094507	-32.840909
35	1.000000	0.785398	-24.923841	-36.108417
36	1.000000	0.785398	-20.389729	-38.849383
37	1.000000	0.785398	-15.558289	-41.023838
38	1.000000	0.785398	-10.499975	-42.600072
39	1.000000	0.785398	-5.288547	-43.555102
40	1.000000	0.785398	0.00000	-43.875000
41	1.000000	0.785398	5.288547	-43.555102
42	1.000000	0.785398	10.499975	-42.600072
43	1.000000	0.785398	15.558289	-41.023838
44	1.000000	0.785398	20.389729	-38.849383
45	1.000000	0.785398	24.923841	-36.108417
46	1.000000	0.785398	29.094507	-32.840909
47	1.000000	0.785398	32.840909	-29.094507
48	1.000000	0.785398	36.108417	-24.923841
49	1.000000	0.785398	38.849383	-20.389729
50	1.000000	0.785398	41.023838	-15.558289
51	1.000000	0.785398	42.600072	-10.499975
52	1.000000	0.785398	43.555102	-5.288547

NOTE: The positions of the above rebars were computed by LPile

Minimum spacing between any two bars not equal to zero = 4.298 inches
between bars 48 and 49.

Ratio of bar spacing to maximum aggregate size = 5.73

Concrete Properties:

Compressive Strength of Concrete = 4500. psi

Modulus of Elasticity of Concrete	=	3823676. psi
Modulus of Rupture of Concrete	=	-503.115295 psi
Compression Strain at Peak Stress	=	0.002001
Tensile Strain at Fracture of Concrete	=	-0.0001152
Maximum Coarse Aggregate Size	=	0.750000 in

Number of Axial Thrust Force Values Determined from Pile-head Loadings = 2

Number	Axial Thrust Force kips
1	58.210
2	93.587

Summary of Results for Nominal Moment Capacity for Section 1

Moment values interpolated at maximum compressive strain = 0.003
or maximum developed moment if pile fails at smaller strains.

Load No.	Axial Thrust kips	Nominal Mom. Cap. in-kip	Max. Comp. Strain
1	58.210	104151.110	0.00300000
2	93.587	105468.258	0.00300000

Note that the values of moment capacity in the table above are not
factored by a strength reduction factor (phi-factor).

In ACI 318, the value of the strength reduction factor depends on whether
the transverse reinforcing steel bars are tied hoops (0.65) or spirals (0.75).

The above values should be multiplied by the appropriate strength reduction
factor to compute ultimate moment capacity according to ACI 318,
or the value required by the design standard being followed.

The following table presents factored moment capacities and corresponding
bending stiffnesses computed for common resistance factor values used for
reinforced concrete sections.

Axial Load No.	Resist. Factor	Nominal Ax. Thrust kips	Nominal Moment Cap in-kips	Ult. (Fac) Ax. Thrust kips	Ult. (Fac) Moment Cap in-kips	Bend. Stiff. at Ult Mom kip-in^2
1	0.65	58.210000	104151.	37.836500	67698.	2.5468E+09
2	0.65	93.586667	105468.	60.831333	68554.	2.5844E+09
1	0.75	58.210000	104151.	43.657500	78113.	2.4547E+09
2	0.75	93.586667	105468.	70.190000	79101.	2.4914E+09
1	0.90	58.210000	104151.	52.389000	93736.	1.5681E+09
2	0.90	93.586667	105468.	84.228000	94921.	1.5948E+09

Layering Correction Equivalent Depths of Soil & Rock Layers

Layer No.	Top of Layer Below Pile Head ft	Equivalent Top Depth Below Grnd Surf ft	Same Layer Type As Layer Above	Layer is Rock or is Below Rock Layer	F0 Integral for Layer lbs	F1 Integral for Layer lbs
1	0.5000	0.00	N.A.	No	0.00	14957.
2	2.5000	1.7107	Yes	No	14957.	61479.
3	4.5000	3.7952	Yes	No	76436.	111679.
4	6.5000	6.6747	Yes	No	188115.	137011.
5	8.5000	8.4400	Yes	No	325126.	527601.
6	13.5000	30.4533	No	No	852727.	179940.
7	18.5000	41.5821	No	No	1032667.	144060.
8	23.5000	16.5378	No	No	1176727.	298662.
9	25.5000	12.8613	No	No	1475389.	272823.
10	27.5000	27.0000	No	No	1748212.	N.A.

Notes: The F0 integral of Layer n+1 equals the sum of the F0 and F1 integrals
for Layer n. Layering correction equivalent depths are computed only

for soil types with both shallow-depth and deep-depth expressions for peak lateral load transfer. These soil types are soft and stiff clays, non-liquefied sands, and cemented c-phi soil.

Summary of Pile-head Responses for Conventional Analyses

Definitions of Pile-head Loading Conditions:

Load Type 1: Load 1 = Shear, V, lbs, and Load 2 = Moment, M, in-lbs
Load Type 2: Load 1 = Shear, V, lbs, and Load 2 = Slope, S, radians
Load Type 3: Load 1 = Shear, V, lbs, and Load 2 = Rot. Stiffness, R, in-lbs/rad.
Load Type 4: Load 1 = Top Deflection, y, inches, and Load 2 = Moment, M, in-lbs
Load Type 5: Load 1 = Top Deflection, y, inches, and Load 2 = Slope, S, radians

Load Case No.	Load Type 1	Pile-head Load 1	Load Type 2	Pile-head Load 2	Axial Loading lbs	Pile-head Deflection inches	Pile-head Rotation radians	Max Shear in Pile lbs	Max Moment in Pile in-lbs
1	V, lb	64747.	M, in-lb	1.03E+08	93587.	6.5793	-0.05242	-588166.	1.06E+08
2	V, lb	13220.	M, in-lb	2.09E+07	58210.	0.1790	-9.27E-04	-112579.	2.13E+07

Maximum pile-head deflection = 6.5792756346 inches
Maximum pile-head rotation = -0.0524226779 radians = -3.003598 deg.

The analysis ended normally.

2020 New York State Building Code 1807.3.2.1

Moment (ft·k)	6,428.91	
Shear (k)	48.56	
Caisson diameter (ft)	8	
Caisson height above ground (ft)	0.5	
Caisson height below ground (ft)	27	
Lateral soil pressure (lb/ft^2)	**383.33**	
Ground to application of force, h (ft)	132.89	
Applied lateral force, P (lb)	48,560	
Lateral soil bearing pressure, S_1 (lb/ft)	3,450.00	
Diameter, b (ft)	8	
A	4.12	$= (2.34P)/(S_1 b)$
Minimum depth of embedment, d (ft)	26.57	$= 0.5A[\,1 + (\,1 + (\,4.36h\,/\,A\,)\,)^{1/2}\,]$